LEARNER GUIDE

One number, many representations

Place Value · Lesson 8. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

One number can be shown in many ways. 3.45 is “three point four five”, a chart with 3 ones, 4 tenths and 5 hundredths, a point on a line between 3 and 4, **£3.45 = 345p** and **3.45 m = 345 cm**. Pence are hundredths of a pound and centimetres are hundredths of a metre, so the digits keep their order: only the unit changes. Each representation shows something best.

What you are learning to notice

- £1 = 100p and 1 m = 100 cm: pence and centimetres are both hundredths.
- To change pounds to pence, or metres to centimetres, the digits move two columns to the left (× 100).
- Money is written with two digits after the point: £3.50, not £3.5.
- A 0 can matter: £2.07 = 207p, but £2.70 = 270p.
- A number line shows size; a chart shows what each digit is worth; words show how to say it.

Moving through it, one step at a time

To change pounds to pence (or metres to centimetres), multiply by 100: every digit moves two columns to the left. To change pence to pounds (or centimetres to metres), divide by 100. Check with a chart: pounds or metres are the ones; pence or centimetres are the hundredths.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — £3.45 in pence

How many pence is £3.45?

£3.45

Ones (£1)		Tenths (10p)	Hundredths (1p)
3	•	4	5

Pence are hundredths of a pound: the 5 is 5p, the 4 is 40p, the 3 is 300p.

1. 3 pounds is 300p
2. 4 tenths of a pound is 40p
3. 5 hundredths of a pound is 5p
4. Altogether 345p

£3.45 = 345p. The digits keep their order; the unit changes.

Example 2 — 1.5 m in centimetres

How many centimetres is 1.5 m?

1.5 m

Ones (1 m)		Tenths (10 cm)	Hundredths (1 cm)
1	•	5	0

1.5 m = 1.50 m. The 0 holds the hundredths (centimetres) column.

1. 1 metre is 100 cm
2. 5 tenths of a metre is 50 cm
3. Altogether 150 cm

1.5 m = 150 cm, not 15 cm.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the whole Place Value unit (Lessons 1 to 7)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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